

Mathematical Modeling and Controller Design for a BlueROV2-Inspired 4-DOF Underwater Vehicle

1. Mathematical Model of the BlueROV2-Inspired AUV/ROV

A complete underwater vehicle has six degrees of freedom: surge, sway, heave, roll, pitch, and yaw. However, for this project, a reduced **4-DOF model** is selected because the main control objective is trajectory and depth tracking. Therefore, the controlled motions are surge, sway, heave, and yaw, while roll and pitch are assumed to be passively stabilized by buoyancy and restoring moments. This type of reduced marine craft model is commonly derived from the standard marine vehicle dynamics framework introduced by Fossen [1].

The generalized position vector is defined in the earth-fixed frame as

$$\eta = \begin{bmatrix} x \\ y \\ z \\ \psi \end{bmatrix} \quad (1)$$

where x and y are horizontal positions, z is depth, and ψ is yaw angle. The body-fixed velocity vector is

$$v = \begin{bmatrix} u \\ v \\ w \\ r \end{bmatrix} \quad (2)$$

where u is surge velocity, v is sway velocity, w is heave velocity, and r is yaw rate.

The reason for using two coordinate frames is that the vehicle position is measured in the earth-fixed frame, while the velocities generated by the thrusters are naturally expressed in the body-fixed frame. Therefore, a coordinate transformation is required.

The kinematic model is

$$\dot{\eta} = J(\psi)v \quad (3)$$

where

$$J(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 & 0 \\ \sin \psi & \cos \psi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiplying the matrix gives

$$\begin{aligned}
\dot{x} &= u \cos \psi - v \sin \psi \\
\dot{y} &= u \sin \psi + v \cos \psi \\
\dot{z} &= w \\
\dot{\psi} &= r
\end{aligned} \tag{4}$$

These equations show that surge and sway velocities must be rotated by the yaw angle before they become earth-frame position rates.

The general marine craft dynamic model is

$$M\dot{v} + C(v)v + D(v)v + g(\eta) = \tau + \tau_d \tag{5}$$

where M is the inertia matrix, $C(v)v$ represents Coriolis and centripetal terms, $D(v)v$ represents hydrodynamic damping, $g(\eta)$ represents restoring forces, τ is the control input, and τ_d is the external disturbance. This form is the standard basis for marine craft modeling and control [1].

For a low-speed BlueROV2-inspired model, the Coriolis and coupling terms are neglected in the first version. Unknown effects are grouped into a disturbance term $\Delta(t)$. Therefore, the simplified model becomes

$$M\dot{v} + Dv + \Delta(t) = \tau \tag{6}$$

where

$$M = \begin{bmatrix} m_u & 0 & 0 & 0 \\ 0 & m_v & 0 & 0 \\ 0 & 0 & m_w & 0 \\ 0 & 0 & 0 & I_z \end{bmatrix}$$

and

$$D = \begin{bmatrix} d_u & 0 & 0 & 0 \\ 0 & d_v & 0 & 0 \\ 0 & 0 & d_w & 0 \\ 0 & 0 & 0 & d_r \end{bmatrix}$$

Here m_u, m_v, m_w are effective masses including added mass effects, and I_z is the effective yaw inertia. The damping coefficients d_u, d_v, d_w, d_r are positive constants representing hydrodynamic resistance.

The disturbance vector is

$$\Delta(t) = \begin{bmatrix} \Delta_u(t) \\ \Delta_v(t) \\ \Delta_w(t) \\ \Delta_r(t) \end{bmatrix} \quad (7)$$

and the control vector is

$$\tau = \begin{bmatrix} \tau_u \\ \tau_v \\ \tau_w \\ \tau_r \end{bmatrix} \quad (8)$$

Expanding the dynamic equation gives

$$\begin{aligned} m_u \dot{u} + d_u u + \Delta_u &= \tau_u \\ m_v \dot{v} + d_v v + \Delta_v &= \tau_v \\ m_w \dot{w} + d_w w + \Delta_w &= \tau_w \\ I_z \dot{r} + d_r r + \Delta_r &= \tau_r \end{aligned}$$

Solving for the acceleration terms:

$$\begin{aligned} \dot{u} &= \frac{1}{m_u} (\tau_u - d_u u - \Delta_u) \\ \dot{v} &= \frac{1}{m_v} (\tau_v - d_v v - \Delta_v) \\ \dot{w} &= \frac{1}{m_w} (\tau_w - d_w w - \Delta_w) \\ \dot{r} &= \frac{1}{I_z} (\tau_r - d_r r - \Delta_r) \end{aligned} \quad (9)$$

Therefore, the full nonlinear state-space model is

$$X = \begin{bmatrix} x \\ y \\ z \\ \psi \\ u \\ v \\ w \\ r \end{bmatrix} \quad (10)$$

$$\dot{X} = \begin{bmatrix} u \cos \psi - v \sin \psi \\ u \sin \psi + v \cos \psi \\ w \\ r \\ \frac{1}{m_u}(\tau_u - d_u u - \Delta_u) \\ \frac{1}{m_v}(\tau_v - d_v v - \Delta_v) \\ \frac{1}{m_w}(\tau_w - d_w w - \Delta_w) \\ \frac{1}{I_z}(\tau_r - d_r r - \Delta_r) \end{bmatrix} \quad (11)$$

This is the mathematical model used for all controller designs.

2. Control Objective

The objective is to make the BlueROV2-inspired vehicle follow a desired 4-DOF trajectory:

$$\eta_d(t) = \begin{bmatrix} x_d(t) \\ y_d(t) \\ z_d(t) \\ \psi_d(t) \end{bmatrix} \quad (12)$$

The position tracking error is

$$e_\eta = \eta - \eta_d \quad (13)$$

The velocity tracking error is

$$e_v = v - v_d \quad (14)$$

The control objective is

$$\lim_{t \rightarrow \infty} e_\eta(t) = 0 \quad (15)$$

or practically,

$$\| e_\eta(t) \| < \epsilon \quad (16)$$

where ϵ is a small acceptable tracking bound.

The set of controllers considered in this study is intentionally selected to represent increasing levels of model knowledge, robustness, and adaptability.

The state feedback controller serves as a baseline linear control strategy, relying on a linearized approximation of the system dynamics. The observer-based controller extends this framework by addressing the practical limitation of incomplete state measurements.

Feedback linearization introduces nonlinear model-based compensation, allowing the system dynamics to be transformed into a linear form under ideal modeling assumptions. The MRAC controller further enhances adaptability by adjusting controller parameters online to cope with uncertain system dynamics.

Finally, the backstepping controller is adopted as the primary nonlinear control strategy due to its compatibility with the cascaded structure of underwater vehicle dynamics and its strong Lyapunov-based stability guarantees.

In addition, an Inverse Neural Network (INN) controller is introduced as a learning-based extension, trained using the backstepping controller as a supervisory signal.

3. Controller 1: State Feedback Control

State feedback is the simplest baseline controller. It is usually designed for a linear state-space system of the form

$$\dot{X} = AX + B\tau \quad (17)$$

where the state vector is

$$X = [x \ y \ z \ \psi \ u \ v \ w \ r]^T \in \mathbb{R}^8$$

and the control input vector is

$$\tau = [\tau_u \ \tau_v \ \tau_w \ \tau_r]^T \in \mathbb{R}^4.$$

Therefore,

$$A \in \mathbb{R}^{8 \times 8}, B \in \mathbb{R}^{8 \times 4}.$$

The matrix A describes the natural dynamics of the vehicle without control input. Its upper part connects position derivatives to body velocities, while its lower part represents damping effects on surge, sway, heave, and yaw motion. The matrix B describes how each control input affects the corresponding acceleration channel. Since forces do not directly change position but first change velocity, the first four rows of B are zero.

This controller is widely used in modern control because it directly places or shapes the closed-loop dynamics using the gain matrix K [2].

Because the BlueROV2 model contains nonlinear terms $\sin \psi$ and $\cos \psi$, we first linearize around a nominal operating point. A common choice is

$$\psi \approx 0$$

At this operating point,

$$\cos 0 = 1, \sin 0 = 0$$

so the kinematics become

$$\begin{aligned}\dot{x} &= u \\ \dot{y} &= v \\ \dot{z} &= w \\ \dot{\psi} &= r\end{aligned}$$

The dynamics become

$$\begin{aligned}\dot{u} &= -\frac{d_u}{m_u}u + \frac{1}{m_u}\tau_u \\ \dot{v} &= -\frac{d_v}{m_v}v + \frac{1}{m_v}\tau_v \\ \dot{w} &= -\frac{d_w}{m_w}w + \frac{1}{m_w}\tau_w \\ \dot{r} &= -\frac{d_r}{I_z}r + \frac{1}{I_z}\tau_r\end{aligned}$$

Therefore,

$$\dot{X} = AX + B\tau$$

where

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\frac{d_u}{m_u} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{d_v}{m_v} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{d_w}{m_w} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{d_r}{I_z} \end{bmatrix}$$

and

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{m_u} & 0 & 0 & 0 \\ 0 & \frac{1}{m_v} & 0 & 0 \\ 0 & 0 & \frac{1}{m_w} & 0 \\ 0 & 0 & 0 & \frac{1}{I_z} \end{bmatrix}$$

For tracking, define

$$e_X = X - X_d$$

The state feedback control law is

$$\tau = -K e_X$$

where $K \in \mathbb{R}^{4 \times 8}$.

A practical diagonal-like version is

$$\begin{aligned} \tau_u &= -k_x(x - x_d) - k_u(u - u_d) \\ \tau_v &= -k_y(y - y_d) - k_v(v - v_d) \\ \tau_w &= -k_z(z - z_d) - k_w(w - w_d) \\ \tau_r &= -k_\psi(\psi - \psi_d) - k_r(r - r_d) \end{aligned}$$

Substituting $\tau = -KX$ into the linear model gives

$$\begin{aligned} \dot{X} &= AX - BKX \\ \dot{X} &= (A - BK)X \end{aligned}$$

The closed-loop system is stable if

$$\text{Re}(\lambda_i(A - BK)) < 0 \tag{18}$$

This means all eigenvalues of $A - BK$ must lie in the left half of the complex plane.

Proof of Lyapunov can also be given. Let

$$A_c = A - BK \tag{19}$$

If A_c is Hurwitz, then for any $Q = Q^T > 0$, there exists $P = P^T > 0$ satisfying

$$A_c^T P + P A_c = -Q \quad (20)$$

Choose

$$V = X^T P X$$

Then

$$\begin{aligned} \dot{V} &= X^T (A_c^T P + P A_c) X \\ \dot{V} &= -X^T Q X < 0 \end{aligned}$$

for $X \neq 0$. Therefore, the closed-loop system is asymptotically stable.

State feedback is useful as a baseline, but it has limitations. It assumes a linearized model, requires full state measurement, and does not explicitly handle nonlinear kinematics or disturbances.

4. Controller 2: Observer-Based Control

In practice, not every state vector is measured directly. BlueROV2 may measure position, depth, and yaw using sensors, but body-frame velocities u, v, w, r may need to be estimated. This motivates observer-based control. The observer structure follows classical state estimation theory from modern control [2].

Assume the measured output is

$$Y = C X \quad (21)$$

where

$$Y = [x \quad y \quad z \quad \psi]^T \in \mathbb{R}^4.$$

Since the full state has eight elements and the measured output has four elements,

$$C \in \mathbb{R}^{4 \times 8}.$$

For position, depth, and yaw measurements,

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

This matrix simply extracts the measured states from the full state vector.

The Luenberger observer is

$$\dot{\hat{X}} = A\hat{X} + B\tau + L(Y - C\hat{X}) \quad (22)$$

where $\hat{X} \in \mathbb{R}^8$ is the estimated state vector and $L \in \mathbb{R}^{8 \times 4}$ is the observer gain matrix.

The term

$$Y - C\hat{X}$$

represents the difference between the measured output and the estimated output. If the estimated position or yaw differs from the measured value, the observer uses this error to correct the estimated states.

Define the estimation error as

$$\tilde{X} = X - \hat{X}$$

The real system is

$$\dot{X} = AX + B\tau$$

The observer is

$$\dot{\hat{X}} = A\hat{X} + B\tau + L(CX - C\hat{X})$$

Subtracting observer from plant:

$$\begin{aligned} \dot{\tilde{X}} &= \dot{X} - \dot{\hat{X}} \\ \dot{\tilde{X}} &= AX + B\tau - (A\hat{X} + B\tau + LC(X - \hat{X})) \end{aligned}$$

Cancel $B\tau$:

$$\begin{aligned} \dot{\tilde{X}} &= A(X - \hat{X}) - LC(X - \hat{X}) \\ \dot{\tilde{X}} &= (A - LC)\tilde{X} \end{aligned}$$

The observer is stable if

$$\text{Re}(\lambda_i(A - LC)) < 0 \quad (23)$$

The control law now uses estimated states:

$$\tau = -K(\hat{X} - X_d) \quad (24)$$

The observer-based controller is more realistic than full-state feedback because it does not require direct measurement of all velocities. However, it still depends on the linearized model and does not directly compensate for nonlinear hydrodynamic uncertainty.

5. Controller 3: Feedback Linearization

Feedback linearization is a nonlinear control method. Its main idea is to cancel known nonlinear dynamics and transform the system into a linear error system. This approach is standard in nonlinear control theory [3].

5.1 Input-State Feedback Linearization

Start with the simplified dynamics without disturbance:

$$M\dot{v} + Dv = \tau \quad (25)$$

The goal is to make the velocity dynamics behave like

$$\dot{v} = a$$

where a is a new artificial input.

Choose

$$\tau = Ma + Dv$$

Substitute this into the system:

$$M\dot{v} + Dv = Ma + Dv$$

Cancel Dv :

$$M\dot{v} = Ma$$

Since M is invertible,

$$\dot{v} = a$$

Now choose

$$a = \dot{v}_d - K_v(v - v_d)$$

Define

$$e_v = v - v_d$$

Differentiate:

$$\dot{e}_v = \dot{v} - \dot{v}_d$$

Since $\dot{v} = a$,

$$\begin{aligned}\dot{e}_v &= \dot{v}_d - K_v e_v - \dot{v}_d \\ \dot{e}_v &= -K_v e_v\end{aligned}$$

If $K_v > 0$, then

$$e_v(t) \rightarrow 0$$

Therefore, the final input-state feedback linearization controller is

$$\tau = M(\dot{v}_d - K_v e_v) + Dv \quad (26)$$

This controller is mainly a velocity tracking controller.

5.2 Input-Output Feedback Linearization

For trajectory tracking, the output is

$$y = \eta \quad (27)$$

The output tracking error is

$$e_\eta = \eta - \eta_d$$

The kinematic model is

$$\dot{\eta} = J(\psi)v$$

We choose a desired body velocity v_d such that the position error decreases. Let

$$\dot{\eta} = \dot{\eta}_d - K_\eta e_\eta$$

But since

$$\dot{\eta} = J(\psi)v$$

we choose

$$J(\psi)v_d = \dot{\eta}_d - K_\eta e_\eta$$

Multiplying by $J^{-1}(\psi)$:

$$v_d = J^{-1}(\psi)(\dot{\eta}_d - K_\eta e_\eta)$$

Now, if $v = v_d$, then

$$\begin{aligned}\dot{e}_\eta &= \dot{\eta} - \dot{\eta}_d \\ \dot{e}_\eta &= J(\psi)v_d - \dot{\eta}_d\end{aligned}$$

Substitute v_d :

$$\begin{aligned}\dot{e}_\eta &= \dot{\eta}_d - K_\eta e_\eta - \dot{\eta}_d \\ \dot{e}_\eta &= -K_\eta e_\eta\end{aligned}$$

Thus, if the inner velocity loop makes $v \rightarrow v_d$, then the outer position loop converges.

The full controller becomes

$$\begin{aligned}v_d &= J^{-1}(\psi)(\dot{\eta}_d - K_\eta e_\eta) \\ \tau &= M(\dot{v}_d - K_v e_v) + Dv\end{aligned}\tag{28}$$

Feedback linearization is stronger than state feedback because it directly handles the nonlinear coordinate transformation. Its weakness is that it assumes accurate knowledge of M and D , so disturbances and parameter uncertainty can degrade performance.

6. Controller 4: MRAC

MRAC is used when the system parameters are uncertain. In underwater vehicles, effective mass and damping may change due to added mass, water conditions, payload changes, or modeling errors. The aim of MRAC is to force the actual system to behave like a desired reference model. [4].

For one velocity channel, for example surge,

$$\dot{u} = au + b\tau_u\tag{29}$$

where a and b may be uncertain.

Define the desired reference model:

$$\dot{u}_m = a_m u_m + b_m u_d\tag{30}$$

where u_m is the ideal model response and u_d is the desired surge velocity command. The parameter $a_m < 0$ is selected so that the reference model is stable.

The objective is

$$u(t) \rightarrow u_m(t)$$

Define the tracking error:

$$e = u - u_m$$

Assume the adaptive control law

$$\tau_u = \theta_1 u + \theta_2 u_d$$

where θ_1 and θ_2 are adaptive parameters.

Substitute into the plant:

$$\begin{aligned}\dot{u} &= au + b(\theta_1 u + \theta_2 u_d) \\ \dot{u} &= (a + b\theta_1)u + b\theta_2 u_d\end{aligned}$$

The ideal parameters θ_1^*, θ_2^* would satisfy

$$\begin{aligned}a + b\theta_1^* &= a_m \\ b\theta_2^* &= b_m\end{aligned}$$

If these ideal parameters were known, the plant would match the reference model exactly. Since they are unknown, MRAC adapts θ_1 and θ_2 online.

A common adaptation law is

$$\begin{aligned}\dot{\theta}_1 &= -\gamma_1 e u \\ \dot{\theta}_2 &= -\gamma_2 e u_d\end{aligned} \tag{31}$$

where $\gamma_1, \gamma_2 > 0$ are adaptation gains.

For the 4-DOF BlueROV2 model, the same idea can be applied independently to each velocity channel:

$$i \in \{u, v, w, r\}.$$

For each channel,

$$\tau_i = \theta_{i1} v_i + \theta_{i2} v_{d,i}.$$

The reference model is

$$\dot{v}_{m,i} = a_{m,i} v_{m,i} + b_{m,i} v_{d,i}.$$

The tracking error is

$$e_i = v_i - v_{m,i}.$$

MRAC is useful because it adapts to uncertain plant parameters. However, it is more sensitive to tuning than feedback linearization and backstepping. If the adaptation gains are too large, the controller can become oscillatory.

To justify the stability of the MRAC scheme, a Lyapunov-based analysis is introduced.

Define the parameter estimation errors as

$$\tilde{\theta}_1 = \theta_1 - \theta_1^*, \tilde{\theta}_2 = \theta_2 - \theta_2^*$$

where θ_1^* and θ_2^* are the ideal parameters that perfectly match the reference model.

Consider the Lyapunov candidate function

$$V = \frac{1}{2}e^2 + \frac{1}{2\gamma_1}\tilde{\theta}_1^2 + \frac{1}{2\gamma_2}\tilde{\theta}_2^2$$

where:

- $e = u - u_m$ is the tracking error
- $\gamma_1, \gamma_2 > 0$ are adaptation gains

This function is **positive definite**, since it is composed of squared terms.

Differentiating V with respect to time:

$$\dot{V} = e\dot{e} + \frac{1}{\gamma_1}\tilde{\theta}_1\dot{\tilde{\theta}}_1 + \frac{1}{\gamma_2}\tilde{\theta}_2\dot{\tilde{\theta}}_2$$

Using the system dynamics and substituting the adaptive laws

$$\dot{\theta}_1 = -\gamma_1 e u, \dot{\theta}_2 = -\gamma_2 e u_d$$

Substituting the plant dynamics and adaptive laws into $\dot{V}$, the parameter-dependent terms cancel explicitly, yielding $\dot{V} = -\alpha e^2$, where $\alpha > 0$.

Since $\dot{V} \leq 0$ the system is **stable in the sense of Lyapunov**, and the tracking error e is bounded and converges to zero.

7. Controller 5: Backstepping Control

Backstepping is highly suitable for underwater vehicles because the model has a cascaded structure:

$$\eta \rightarrow v \rightarrow \tau$$

This means position depends on velocity, and velocity depends on force. Backstepping is designed exactly for such systems and is commonly derived using Lyapunov stability theory [3].

The kinematic equation is

$$\dot{\eta} = J(\psi)v$$

The dynamic equation is

$$M\dot{v} + Dv = \tau$$

First define position error:

$$e_\eta = \eta - \eta_d$$

Differentiate:

$$\begin{aligned}\dot{e}_\eta &= \dot{\eta} - \dot{\eta}_d \\ \dot{e}_\eta &= J(\psi)v - \dot{\eta}_d\end{aligned}$$

The first step of backstepping is to design a virtual velocity command v_d . Choose

$$v_d = J^{-1}(\psi)(\dot{\eta}_d - K_\eta e_\eta),$$

where

$$K_\eta = K_\eta^T > 0.$$

If the actual velocity satisfies $v = v_d$, then

$$\dot{e}_\eta = J(\psi)v_d - \dot{\eta}_d.$$

Substituting v_d ,

$$\dot{e}_\eta = \dot{\eta}_d - K_\eta e_\eta - \dot{\eta}_d.$$

Therefore,

$$\dot{e}_\eta = -K_\eta e_\eta.$$

This shows that the position error decreases exponentially if the velocity follows v_d which guarantees exponential convergence of the position error.

However, v cannot be directly commanded. The real control input is τ . Therefore, define velocity error:

$$e_v = v - v_d$$

The velocity dynamics are

$$M\dot{v} + Dv = \tau$$

Differentiate velocity error:

$$\dot{e}_v = \dot{v} - \dot{v}_d$$

From the dynamics,

$$\dot{v} = M^{-1}(\tau - Dv)$$

Choose the actual control input:

$$\tau = M(\dot{v}_d - K_v e_v) + Dv \tag{32}$$

where

$$K_v = K_v^T > 0.$$

Substitute into the dynamics:

$$M\dot{v} + Dv = M(\dot{v}_d - K_v e_v) + Dv$$

Cancel Dv :

$$M\dot{v} = M(\dot{v}_d - K_v e_v)$$

Since M is nonsingular,

$$\dot{v} = \dot{v}_d - K_v e_v$$

Then differentiate the velocity error and substituting $\dot{v}$:

$$\begin{aligned} \dot{e}_v &= \dot{v} - \dot{v}_d \\ \dot{e}_v &= -K_v e_v \end{aligned}$$

Thus, the velocity error converges to zero.

Therefore, under the selected control law, both error dynamics are

$$\begin{aligned}\dot{e}_\eta &= -K_\eta e_\eta, \\ \dot{e}_v &= -K_v e_v.\end{aligned}$$

Now choose the Lyapunov function

$$V = \frac{1}{2} e_\eta^T e_\eta + \frac{1}{2} e_v^T M e_v.$$

Because $M = M^T > 0$, the Lyapunov function is positive definite:

$$V > 0 \text{ for } (e_\eta, e_v) \neq 0.$$

Differentiate:

$$\dot{V} = e_\eta^T \dot{e}_\eta + e_v^T M \dot{e}_v.$$

Substitute the closed-loop error dynamics:

$$\dot{V} = e_\eta^T (-K_\eta e_\eta) + e_v^T M (-K_v e_v).$$

Thus,

$$\dot{V} = -e_\eta^T K_\eta e_\eta - e_v^T M K_v e_v.$$

Since $K_\eta > 0$, $K_v > 0$, and $M > 0$,

$$\dot{V} < 0$$

for all nonzero errors. Therefore,

$$e_\eta \rightarrow 0, e_v \rightarrow 0.$$

This proves asymptotic convergence of both position and velocity tracking errors under the selected backstepping control law.

8. Advanced Extension: Independent INN Trained by Backstepping

The Inverse Neural Network controller is introduced as an additional learning-based controller. Neural networks are useful because they can approximate nonlinear input-output mappings, which is important when the exact inverse dynamics of the underwater vehicle are difficult to model [5].

In this project, the expert controller is the backstepping controller. Therefore, backstepping is used as a teacher, and the INN is used as a student.

Training Mode

During training, backstepping generates the target control input:

$$\tau_{BS} = M(\dot{v}_d - K_v e_v) + Dv$$

The INN learns the mapping

$$X_{NN} \rightarrow \tau_{BS}$$

where

$$X_{NN} = \begin{bmatrix} e_\eta \\ e_v \\ v_d \\ v \end{bmatrix}$$

For the 4-DOF model, $e_\eta \in \mathbb{R}^4$, $e_v \in \mathbb{R}^4$, $v_d \in \mathbb{R}^4$, $v \in \mathbb{R}^4$. Therefore, $X_{NN} \in \mathbb{R}^{16}$.

The INN output is

$$\tau_{INN} = f_{NN}(X_{NN}) \tag{33}$$

The training loss is

$$J_{NN} = \frac{1}{2} \| \tau_{BS} - \tau_{INN} \|^2 \tag{34}$$

The objective of training is to minimize J_{NN} , so that $\tau_{INN} \approx \tau_{BS}$.

Testing Mode

After training is completed, the backstepping controller is removed from the control loop. The INN then operates independently.

During testing, the control input is generated directly by the INN:

$$\tau = \tau_{INN}.$$

This means the vehicle is controlled by the learned inverse model:

$$(e_\eta, e_v, v_d, v) \rightarrow \tau_{INN}.$$

The purpose of this comparison is to evaluate whether the INN can reproduce the expert behavior of the backstepping controller while acting as an independent controller.

This progression enables a systematic comparison between linear, nonlinear, adaptive, and learning-based control strategies within a unified framework.

9. Optional Thesis Extension: Fault-Tolerant Allocation

For the thesis, the generalized force command τ_c generated by backstepping or adaptive backstepping can be converted into individual thruster commands using control allocation. Control allocation is especially important for over-actuated vehicles such as BlueROV2, where multiple thrusters can generate the same generalized force [6].

The relation between thruster inputs and generalized force is

$$\tau = B\Lambda u$$

where B is the thruster configuration matrix, u is the thruster command vector, and Λ is the actuator effectiveness matrix.

If one thruster loses effectiveness, Λ changes. The allocator solves

$$u^* = \arg \min_u \|W_\tau(B\Lambda u - \tau_c)\|^2 + \lambda_w \|W_u u\|^2$$

subject to

$$u_{\min} \leq u \leq u_{\max}$$

Simulation and Implementation Framework

The above controllers will be implemented and evaluated within a unified simulation framework based on the derived nonlinear state-space model of the BlueROV2-inspired vehicle.

The simulation environment will be incorporated:

- the full nonlinear kinematic and dynamic equations,
- time-varying disturbance inputs,
- and identical trajectory tracking tasks for all controllers.

This unified framework ensures a fair and consistent comparison of controller performance in terms of trajectory tracking accuracy, control effort, and robustness against disturbances.

References

- [1] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., Wiley, 2021.
- [2] K. Ogata, *Modern Control Engineering*, 5th ed., Pearson/Prentice Hall, 2010.
- [3] J.-J. E. Slotine and W. Li, *Applied Nonlinear Control*, Prentice Hall, 1991.
- [4] P. A. Ioannou and J. Sun, *Robust Adaptive Control*, Prentice Hall, 1996.
- [5] S. Haykin, *Neural Networks and Learning Machines*, 3rd ed., Pearson, 2009.
- [6] T. A. Johansen and T. I. Fossen, “Control allocation—A survey,” *Automatica*, vol. 49, no. 5, pp. 1087–1103, 2013.